

Perceptron

The perceptron is simplest unit (a single artificial neuron)

It was invented by Frank Rosenblatt in 1958. Closely inspired by how a biological neuron fires.

Parts of the perceptron:

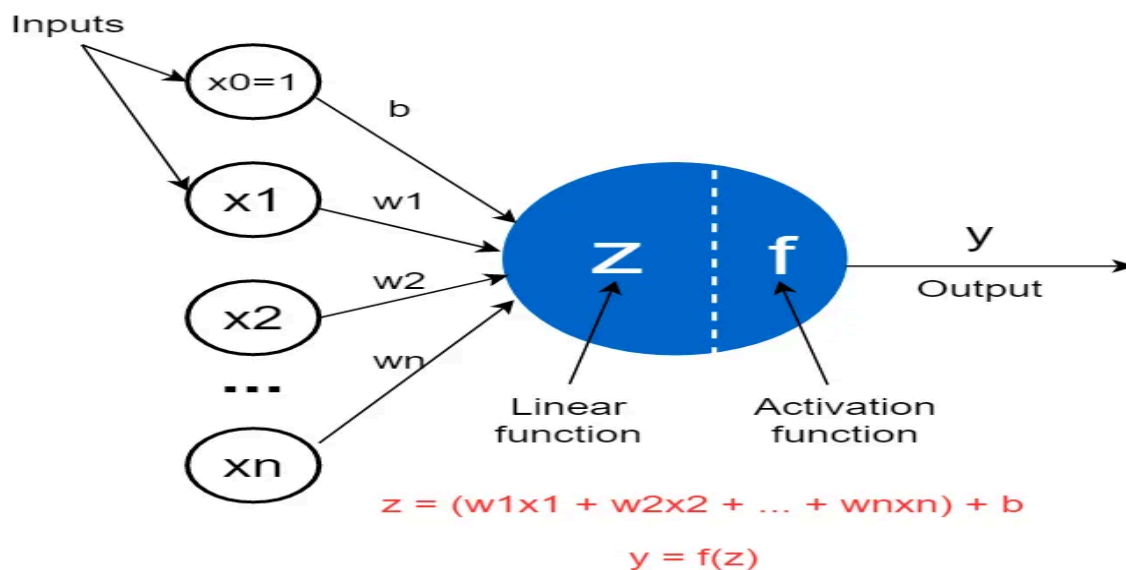
01. Input($x_1, x_2, x_3 \dots x_n$): The features you feed in perceptron. Facial expression, tail length, color texture ..etc.

02. Weights($w_1, w_2, w_3, \dots w_n$): Each input get a weight that says how important the input is. Large weight input strongly influences the decision.

Near zero means input barely matters. Weight are what the perceptron learns.

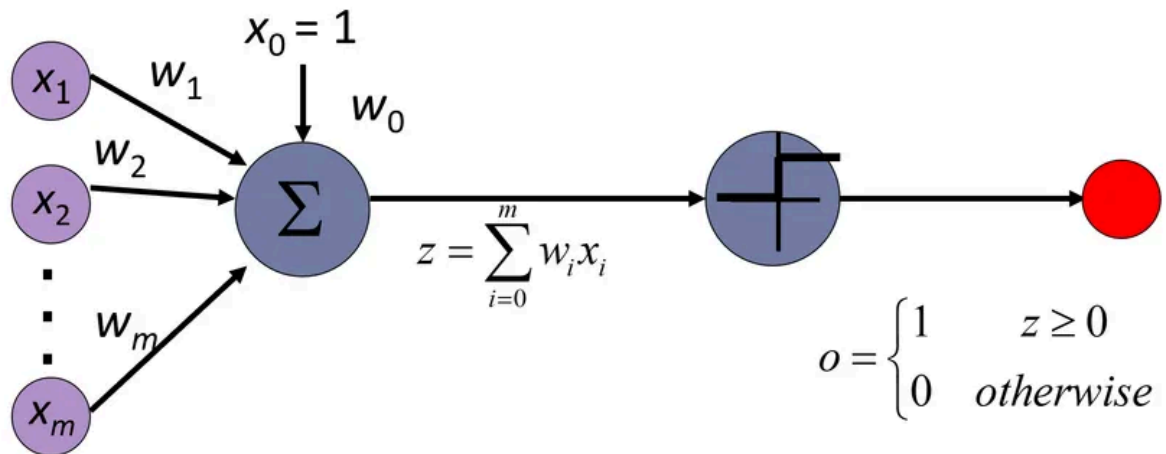
03. Weighted Sum: The perceptron multiplies each input by its weights and adds them all up:

$$Z = x_1w_1 + x_2w_2 + x_3w_3 + \dots + x_nw_n$$



04. Bias(b): An extra adjustable number added to the sum. It shifts the threshold up or down making the neuron easier or harder to activate.

05. Step function: The sum (z) goes through a function that decides the output. In the classic perceptron this is a step function: if z is above 0, output 1; otherwise output 0.



Question: Take the decision should i go outside or not. Inputs are temperature, Raining, Work Status. On the based on input i want to decide .

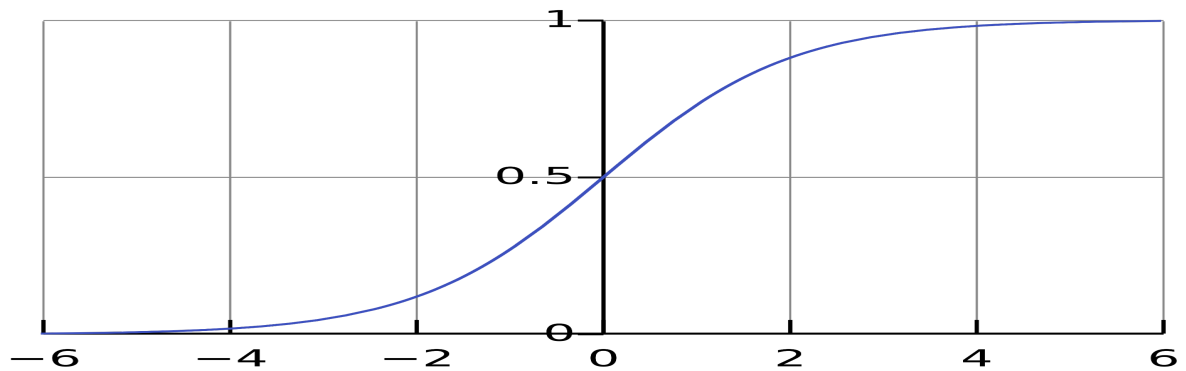
Temperature = 1, $w_1 = 0.6$

Raining = 1, $w_2 = 0.3$

Work Status = 1(i am free), $w_3 = 0.45$

$$Z = 0.6 + 0.3 + 0.45 = 1.35$$

Go to the step function (sigmoid function):



Output = 1 (you should go outside)